A Geometric Criterion for Sway and Rocking in Inclined Pile Groups, with Field Identification of an Offshore Wind Turbine Foundation

Nguyen Hoang Bui[1], Thi Cam Nhung Nguyen*[2], Huu-Quyet Nguyen[3], Chung Truong Thanh[4], Hung Nguyen-Dang[5], and José C. Matos[6]

[1]Department of Mechanical and Aerospace Engineering, University at Buffalo, The State University of New York, Buffalo, NY 14260, USA
[2]University of Transport and Communications, No. 3 Cau Giay Street, Hanoi, Vietnam
[3]The University of Transport and Communications Limited Company, Vietnam
[4]Nha Trang University, 02 Nguyen Dinh Chieu, Khanh Hoa, Vietnam
[5]Professeur Honoraire Ordinaire de l'Université de Liège, Liège, Belgium
[6]Department of Civil Engineering, University of Minho, Campus de Azurém, 4800-058 Guimarães, Portugal

Abstract

Many nearshore wind turbines stand on an elevated concrete cap carried by inclined piles, yet field vibration measurements of such foundations are few. The most useful result of such a measurement is the fundamental mode, but two things must be known before it can be used: whether the mode is a sway or a rocking of the cap, and whether the identified candidate mode is a real mode at all. This paper settles both without knowing the soil stiffness. The axes of outward-raked piles meet near a focal point high above the cap, and the cap can rotate about that point without stretching any pile. The fundamental mode is therefore a sway, and the pile layout bounds its vertical motion on the cap to less than a tenth of its horizontal motion, whereas rocking moves the cap vertically about as much as horizontally. Three checks derived from the definition of a mode then separate real modes from electrical interference and local sensor responses. The method is applied to the 34-pile foundation of turbine T18 of the Dong Hai 1 wind farm in Vietnam, measured before and after tower installation. Only one band, near 1.2 Hz, passes the three checks. Its vertical motion is at most a tenth to a sixth of its horizontal motion, which excludes rocking. Its frequency places the effective fixity of the piles two to five diameters below the seabed, and no change of this frequency larger than about 5 % could be resolved after the tower was installed.

Keywords: inclined pile group; pile-group foundation; operational modal analysis; sway and rocking; offshore wind turbine; soil–structure interaction

1 Introduction

Some offshore and many nearshore wind turbines stand on an elevated pile cap: a concrete cap above the water, carried by a group of inclined piles. It is the foundation of the Donghai Bridge wind farm in China and of the nearshore farms of the Mekong Delta in Vietnam [1–3].

*Corresponding author: ncnhung@utc.edu.vn

Measuring the vibration of the cap checks such a foundation without interrupting its construction or operation. The natural frequencies and mode shapes identified from ambient records can be compared with the design, can serve as a baseline for later monitoring of scour or soil degradation [4, 5], and can be inverted for the soil properties that the design had to assume.

The most useful result of such a measurement is the fundamental mode of the foundation. It is usually the clearest mode in the ambient response, and its stiffness affects the natural frequency of the whole turbine, which must stay away from the rotor and wave frequencies [6]. Two things must be known before this mode can be used. The first is what kind of motion it is. In a sway the cap moves mainly sideways; in a rocking it tilts about a horizontal axis near the pile heads. The two motions are resisted by different parts of the foundation: rocking by the axial stiffness of the piles, sway mainly by their bending and by the soil just below the seabed. A frequency attributed to the wrong motion is inverted for the wrong soil property, and a baseline built on it monitors the wrong part of the foundation. The second is whether the pole taken for this mode is a real mode of the foundation at all. An identification algorithm fits a model to the records and returns poles: candidate modes, each with a frequency, a damping ratio and a shape. Interference from the mains supply and the resonance of a sensor mounting also appear as poles, and they can reappear as consistently from one model order to the next as a structural mode. In the records analysed here, many of the prominent poles are of this kind.

Both questions are usually settled with a numerical model of the foundation. The stiffness of a rigid cap on inclined piles is a classical problem [7–11], and later work extended it to dynamics: the frequency dependence of pile-group impedances [12, 13], the effect of batter on the superstructure [14, 15], experiments on inclined and near-shore piles [16, 17], the complete impedance functions of inclined groups [18, 19], and the coupled horizontal–rocking impedance of an elevated cap [20]. All of these models need the soil stiffness, which is the least certain quantity in the problem: it depends on strain level, on pile installation and on time, and for a foundation measured during construction it is rarely known to better than a factor of two. A classification that rests on the soil model inherits that uncertainty.

Field measurements have not yet supplied an independent check. Output-only identification is well established for offshore wind turbines [4, 5, 21, 22], but separating a structural change from environmental effects requires long records with environmental data [23], and almost all of this work concerns monopiles. Measurements during construction are rare [24, 25], and they tend to lack what model updating relies on: a documented sensor layout, a calibrated model and environmental data. For elevated caps on pile groups, the few studies available describe the modes of the turbine as a whole [26–28]. None measures the motion of the cap itself and compares it with a prediction made beforehand.

Nor do the usual tools decide which poles are modes. Subspace identification [29, 30] and frequency domain decomposition [31] are the established output-only methods [32]. A pole is normally accepted when it stabilises across model orders, but stabilisation tests the consistency of the fit rather than the physics of the pole [33, 34], and automated procedures and uncertainty bounds refine the same idea [35–37]. The question matters beyond modal analysis: neural networks trained on bridge records to reconstruct missing data [38] or detect damage [39] learn only what the records actually contain.

This paper classifies the mode without knowing the soil stiffness, in two steps. The first step uses the geometry of the pile group. When the piles are raked outward at the usual shallow angles, their axes meet close to one point high above the cap. The cap can rotate about that point without stretching or compressing any pile, so the axial stiffness of the piles, and with it most of the soil stiffness, does not resist this motion. The condition is implicit in the classical stiffness matrices [7, 8]; here it is turned into a prediction that can be measured. The fundamental mode is a sway about a centre at or above that point. On the cap it moves almost horizontally, with a vertical component that the pile layout bounds to a small fraction of the horizontal, for a cap whose mass lies near the pile heads; rocking would move the edges of the cap vertically

about as much as horizontally. Comparing the two components measured on the cap therefore tells which motion a mode is. Only the later estimate of the fixity depth of the piles uses the borehole.

The second step applies three checks that any real mode of the foundation must pass, using the measured records and one input from the model, the width of the group of closely spaced modes. A real mode moves the whole cap, so it must appear on several sensors; a pole carried by a single sensor is a local response, such as a resonance of that sensor's mounting [40, 41]. A natural frequency belongs to the structure, not to the sensors, so it must reappear when the sensors are moved, which multi-setup identification assumes rather than tests [42, 43]. And a real structure loses energy as it vibrates; a pole that hardly decays is a steady tone, such as a harmonic of the mains supply [44–46].

The approach is applied to the foundation of turbine T18 of the Dong Hai 1 nearshore wind farm in Vietnam: a 20.5 m cap on 34 piles, 30 of them raked, measured before and after the tower was installed. Only one band of poles, near 1.2 Hz, passes the three checks. On the cap its vertical motion is at most a tenth to a sixth of its horizontal motion, which excludes rocking. Its frequency places the effective fixity of the piles a few diameters below the seabed. After the tower was installed, no change of this frequency larger than about 5 % could be resolved. These results come from a simple campaign, with five or six accelerometers, short records and no environmental data.

Section 2 describes the foundation and the measurements. Section 3 derives the geometric criterion from the pile layout. Section 4 identifies the modes and applies the three checks. Section 5 compares the measured mode with the prediction and discusses the piles and the effect of the tower. The details of the model are given in two appendices, and supporting analyses in the Supplementary Material.

2 The foundation and the field measurements

2.1 Foundation geometry and ground conditions

The foundation carries turbine T18 of the Dong Hai 1 wind farm, Tra Vinh, Vietnam. It consists of a circular reinforced concrete cap, 20.5 m in diameter and 4 m thick, supported on 34 prestressed high-strength concrete piles of 1.0 m outside diameter and 46 m length, arranged in three concentric rings (Table 1 and Figure 1). The inner ring is vertical; the middle and outer rings are raked outward at 7:1 and 5:1. Of this geometry, the prediction of the fundamental mode needs only two quantities: the radius of each ring and the inclination of the piles in it.

Table 1: Foundation geometry and pile properties, from the design drawings. The wall thickness and the elastic modulus of the piles are not given there and are assumed values for this type of pile

Cap	20.5 m diameter, 4 m thick; soffit +5.44 m, top +9.44 m
Inner ring	$R = 3.0$ m, 4 piles at 90°, vertical
Middle ring	$R = 6.0$ m, 12 piles at 30°, inclined 7:1 (8.1° from vertical)
Outer ring	$R = 9.0$ m, 18 piles at 20°, inclined 5:1 (11.3° from vertical)
Pile	prestressed high-strength concrete, 1.0 m outside diameter, 46 m long; toe at ≈ -40.5 m
Free length in water	12.34 m (cap soffit +5.44 m to seabed −6.90 m)
Wall thickness (assumed)	130 mm $\rightarrow A = 0.3553$ m^2, $I = 0.03437$ m^4
Elastic modulus (assumed)	40 GPa $\rightarrow EA = 1.42 \times 10^{10}$ N, $EI = 1.37 \times 10^{9}$ N m^2

The ground is known from borehole DH9 (Figure 2). Below the seabed, a 2.6 m layer of medium dense clayey sand lies over 13.4 m of very soft fat clay, with SPT N not exceeding 2. Stiffer ground begins only 16.0 m below the seabed. Each pile therefore stands free in water

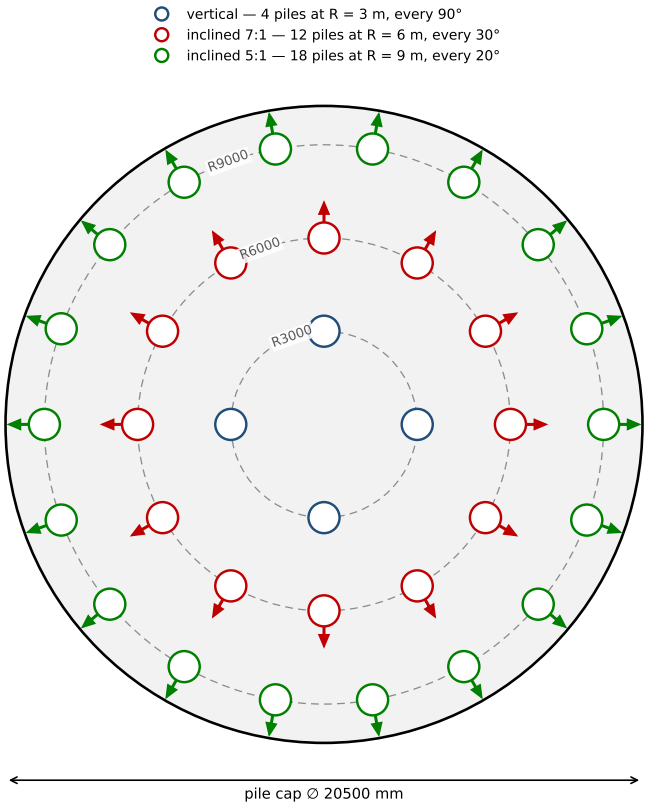


Figure 1: Pile layout of the T18 foundation, redrawn from the setting-out sheet *D1000 PHC pile coordinate*: 34 prestressed high-strength concrete (PHC) piles of 1.0 m diameter on three concentric rings of radius 3.0, 6.0 and 9.0 m, carrying 4, 12 and 18 piles respectively. Arrows give the direction of inclination; the inner ring is vertical, the middle ring is inclined 7:1 and the outer ring 5:1. The cap is 20.5 m in diameter

over 12.34 m and then passes through soft soil before reaching firm ground. This long, weakly supported upper part makes the piles flexible in bending, which is why the fundamental frequency of this foundation lies near 1 Hz.

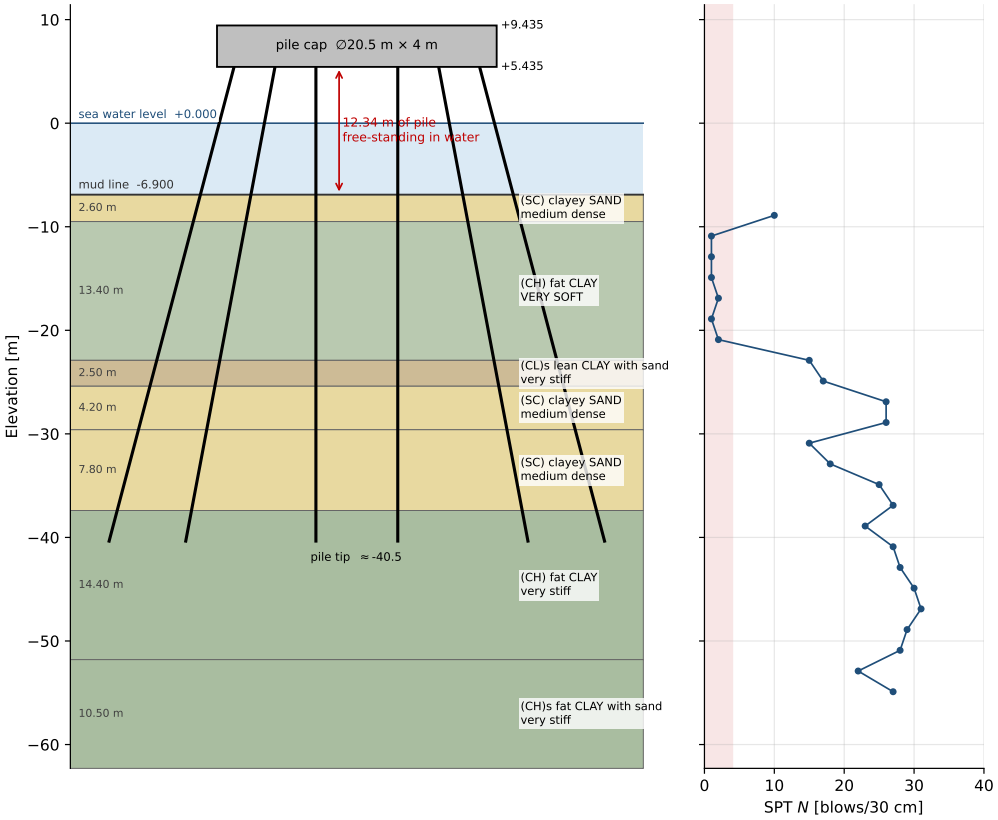


Figure 2: Foundation section and the soil profile at borehole DH9, with the SPT record alongside. Elevations are in metres. Each pile stands free in water over 12.34 m between the cap soffit at +5.44 m and the seabed at −6.90 m. Below a 2.6 m layer of clayey sand, the fat clay is very soft, with $N \leq 2$ over 13.4 m

2.2 Measurement campaigns

Two ambient vibration campaigns were recorded on the top surface of the cap: campaign 1 on 18 May 2024, before the tower was erected, and campaign 2 on 13 July 2024, after. Each was completed within about one and a half hours and comprises several setups that differ in sensor position and measurement direction (Table 2). The sensors sit on a square frame of 5 × 5 m concentric with the cap, at its four corners and at its centre; in campaign 2 a sixth channel was added on the tower, two metres above the centre (Figure 3). The records were acquired with a National Instruments system at 1651.6 Hz. The sensitivities of the accelerometers differ by at most 3 % between channels, so amplitudes can be compared directly from one channel to another. The orientation of the individual channels is recorded only for the lateral setup of campaign 2.

The records are not clean. Single channels show isolated impulsive events, most likely knocks on or near a sensor; most of the signal energy lies in the mains harmonics between 50 and 200 Hz, with less than 1 % in the structural band below 5 Hz; and in campaign 1 the signal level of some channels changes strongly during a record (Supplementary Section S1). Two cautions follow. First, the damping ratios given by the identification are distorted by impulsive events [47], so they are used only to reject poles that hardly decay, not to describe the foundation. Second, the identified frequencies are less precise than their spread within one identification suggests, so poles from different setups are not judged by their frequencies alone.

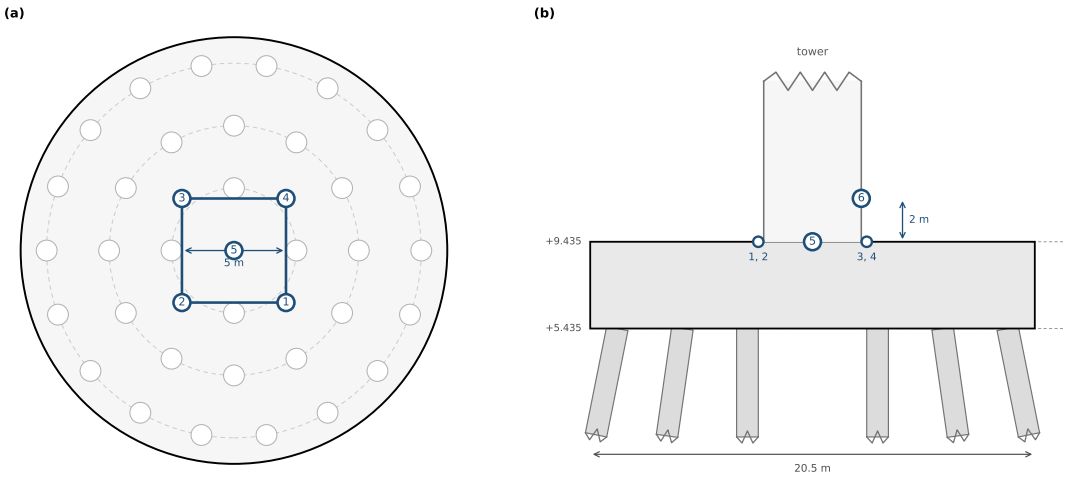


Figure 3: The instrumented frame. (a) Plan of the 5×5 m frame on the 20.5 m cap, with channels 1 to 4 at the corners and channel 5 at the centre. (b) Elevation, showing the sixth channel of campaign 2, which is mounted on the tower two metres above the cap surface. Only the six piles cut by a diametral section are drawn, and both they and the tower are shown broken off rather than at full length

Table 2: Measurement setups. In the lateral setup of campaign 2, channels 1, 2, 3, 5 and 6 measure horizontally in one common direction and channel 4 vertically, as recorded in the processing log. The labels d1 and d2ke are the field names of the two vertical records of campaign 2

Campaign	Setup	Measured direction	Channels	Length (min)
1	Vertical	all vertical	5	25.6
1	Lateral 1	horizontal	5	15.4
1	Lateral 2	horizontal	5	14.0
1	Three-channel	one vertical, two horizontal	3	15.6
2	Lateral	horizontal; channel 4 vertical	6	9.1
2	Vertical d1	all vertical	6	18.9
2	Vertical d2ke	all vertical	6	22.8

3 The geometric criterion for sway and rocking

This section predicts the shape of the fundamental mode from the pile layout. The cap is treated as a rigid body, which is appropriate because it is 4 m thick over 20.5 m while the piles are 1.0 m in diameter over 46 m. The argument is given here for motion in one vertical plane; the full model with six degrees of freedom, used for all numerical values, is given in Appendix A.

3.1 The focal point of an inclined pile group

Figure 4(a) shows the idea on which the paper builds. The axes of the inclined piles are extended upward until they meet at a point F above the cap. If the cap rotates about F, each pile head moves at right angles to the axis of its pile. No pile is then stretched or compressed, so the axial stiffness of the piles, which carries most of the stiffness of the soil, does not resist this motion.

The height of F follows from the elongation of one pile. Let the cap translate horizontally by u and rotate by φ about a horizontal axis through the centre of its soffit, where the pile heads are. A pile with its head at radius r and azimuth θ, measured from the direction of u, and raked outward at α from vertical, changes length by $\cos\alpha\cos\theta\,(u\tan\alpha + r\varphi)$. For a ring of piles with the same radius and inclination this is zero for every pile when $u = -r\varphi\cot\alpha$, which

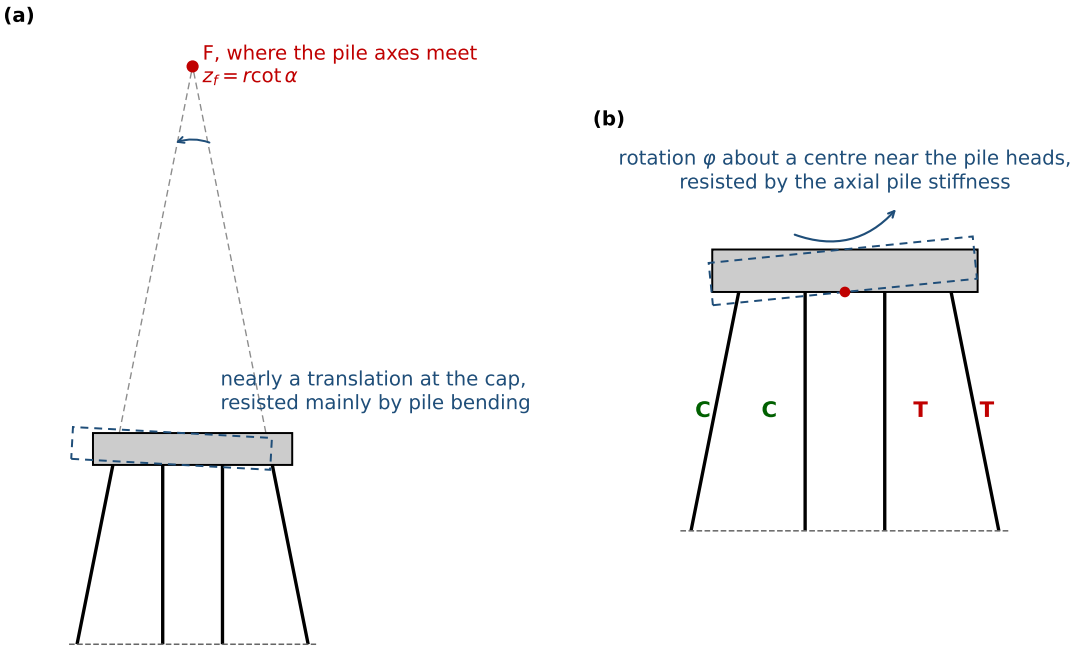


Figure 4: The two lateral mechanisms of a rigid cap on an inclined pile group. (a) Rotation about the focal point F at which the axes of the inclined piles meet, which stretches none of them; at the cap it is nearly a translation, and it is resisted mainly by pile bending. (b) Rocking about a centre near the pile heads, resisted by the axial pile stiffness acting on the lever arms. Dashed outlines give the displaced position; C marks a pile in compression and T a pile in tension. Only one ring of inclined piles is drawn

is a rotation about the point on the cap axis at height

$$z_f = r\cot\alpha \tag{1}$$

above the pile heads: the focal point F of the ring.

In a group with several rings, the focal points need not coincide. In the T18 group they lie close together: 42.0 m above the pile heads for the middle ring and 45.0 m for the outer ring, while the vertical inner ring adds only rotational stiffness. The axial stiffness of the group therefore still leaves one motion almost free, a rotation about a centre 44.2 m above the pile heads, between the two focal points. About that centre the piles provide only 2 % of the axial rotational stiffness they provide about the pile heads, almost all of it from the vertical inner ring (Appendix A).

3.2 Sway, rocking and the criterion

Because the axial stiffness hardly resists rotation about F, that motion is resisted mainly by pile bending and is the fundamental mode of the group. In the rigid-cap model its centre of rotation z_c lies above F for every combination of soil parameters examined (Section 3.3), because the inertia of the cap, which sits at the pile heads, offsets most of the translational stiffness. Seen on the cap, which lies far below F, the mode is almost a sway. A point on the top of the cap, at height h above the pile heads and at horizontal distance d from the rotation axis, moves vertically by $d/(z_c - h)$ of its horizontal motion. With $z_c \geq z_f$, the ratio of the vertical motion w to the horizontal motion u is bounded:

$$\left.\frac{w}{u}\right|_{\text{top}} \leq \frac{d}{z_f - h}. \tag{2}$$

The bound contains only the pile layout, but it holds for a cap whose mass lies near the pile heads; a heavy superstructure can lower the centre of rotation. For T18, $h = 4$ m, and a corner of the instrumented frame lies at most 3.5 m from the rotation axis whatever the direction of sway (2.5 m if the sway is parallel to a side of the frame, 3.5 m if it is along a diagonal), so the vertical motion of the fundamental mode is at most 9 % of its horizontal motion.

Figure 4(b) shows the other lateral motion, rocking about a centre near the pile heads. The piles on one side are compressed and those on the other side stretched, so rocking is resisted by the full axial stiffness and has a much higher frequency. At the top of the cap its vertical motion is about as large as its horizontal motion.

The ratio of vertical to horizontal motion measured on the cap therefore separates sway from rocking by an order of magnitude. This is the geometric criterion.

Torsion, a twist of the cap about its vertical axis, also stretches none of the outward-raked piles, because each pile lies in a radial plane and its head moves at right angles to that plane. Like the fundamental mode, torsion is resisted by pile bending, and its frequency lies close to the fundamental frequency. The fundamental mode is thus expected as a group of closely spaced modes: two lateral modes of equal frequency in perpendicular directions (the fundamental pair), and torsion near them.

3.3 Predictions for the T18 foundation

The pile layout fixes the shapes of these modes but not their frequencies, which depend on the soil through two quantities: the axial stiffness k_{ax} of a pile and its effective fixity length L_{eff}, from the pile head to the point at which it is effectively fixed. The part of this length below the seabed is the depth to fixity. Table 3 gives the predictions of the full model when both are varied over wide ranges: k_{ax} from 1.5 to 6×10^8 N/m, which brackets the values estimated from the soil profile (Appendix B), and the depth to fixity from 1.5 to 12 pile diameters, which covers the values inferred in Section 5.2.

Table 3: Modes of the rigid-cap model of the T18 group over the swept range of axial stiffness (1.5–6×10^8 N/m per pile), depth to fixity (1.5–12 pile diameters) and participating pile mass (Appendix A). The vertical-to-horizontal ratio is evaluated at the cap top for a point 2.5 or 3.5 m from the rotation axis

	Sway (fundamental pair)	Rocking pair
Frequency	0.62–1.53 Hz	5.4–12.2 Hz
Centre of rotation above the pile heads	48–150 m	0.9–1.8 m
Vertical/horizontal at the cap top, $d = 2.5$ m	0.017–0.057	0.8–1.1
Vertical/horizontal at the cap top, $d = 3.5$ m	0.024–0.080	1.1–1.6
Torsion, f_{tor}/f_1	1.08–1.21	
Rocking, f_{rock}/f_1	4.4–16	
Increase of f_1 for a fourfold increase of k_{ax}	5–6 %	

Three predictions matter for what follows. First, the centre of rotation of the fundamental pair stays above the focal points in every case (at least 48 m), so the bound (2) holds over the whole range: in the full model its vertical motion is 2 to 8 % of its horizontal motion, against 80 to 160 % for rocking. Second, the fundamental frequency depends on the fixity length and hardly on the axial stiffness, whereas rocking lies 4 to 16 times higher and rises with the axial stiffness. Third, torsion lies only 8 to 21 % above the fundamental frequency, so the low band of this foundation should contain several poles within about 20 % of each other.

4 Identification of the real modes

4.1 Identification procedure

The modes are identified by covariance-driven subspace identification [29, 30], with the spectral peaks of frequency domain decomposition [31] reported alongside (Supplementary Table S2). Subspace identification builds its model from the covariances of the records over a time window; if this window is much shorter than the period of a mode, the mode cannot be resolved [48]. At the acquisition rate the window is only 0.09 s, a tenth of a period at 1 Hz. The records are therefore decimated to 51.61 Hz and high-pass filtered at 0.3 Hz, which gives a window of 2.33 s with 120 block rows, and identified over model orders 2 to 80 (tested on one record, Supplementary Section S4). Poles with a damping ratio outside 0.1 to 15 % are discarded; in the test at the acquisition rate (Supplementary Section S4) the lower limit is relaxed to 0.001 % so that undamped tones remain visible. Poles from successive orders are grouped when they agree within 1 % in frequency, 5 % in damping and 0.98 in modal assurance criterion, and a group is kept when it contains at least eight poles.

4.2 Three checks from the definition of a mode

Each identified pole is then tested against three checks.

Check 1: a mode moves more than one sensor. A pole that appears on a single channel describes a local motion, such as that of a sensor mounting, not the structure. The spread of a pole over the channels is measured by the effective channel count

$$N_{\mathrm{eff}} = \frac{\left(\sum_j |\phi_j|^2\right)^2}{\sum_j |\phi_j|^4}, \tag{3}$$

the participation ratio of Bell and Dean [49, 50], computed from the first singular vector $\boldsymbol{\phi}$ of the cross-power spectral density matrix. It equals the number of channels when all channels respond equally, and 1 when only one responds. To the authors' knowledge it has not been used before to accept or reject poles in operational modal analysis. It is evaluated once for each setup and band, at the spectral peak. A band is rejected when $N_{\mathrm{eff}} < 1.25$, a value placed between the 1.0 that a single-channel response gives in simulation and the 1.5 of the measured shape (Supplementary Section S2). When the levels of the channels differ strongly the check can reject a real mode, in about a fifth of simulated trials; it never accepted a local response. The threshold is not applied to the three-channel setup, where the two cases cannot be told apart.

Check 2: a mode is the same whichever sensors observe it. Its frequency belongs to the structure, so it must reappear in every setup of a campaign, and where two setups measure the same points, their mode shapes must agree. Because the pile layout predicts a group of closely spaced modes (Section 3.3), the check is applied to the group: every setup able to observe it must return a pole within it, and the group must be no wider than about 20 %. This width is the one input that the checks take from the model.

Check 3: a mode loses energy. A steady harmonic, such as interference from the mains supply, appears to the estimator as a pole with almost no damping [44, 45]; such harmonics can also be recognised from the statistics of the response [51, 52] and removed [46, 53]. Here poles with a damping ratio below 0.1 % are rejected. This bound equals the lowest damping that EN 1991-1-4 gives for any structural element, a logarithmic decrement of 0.006 for parallel cables [54], and lies more than an order of magnitude below the damping of the band finally accepted.

4.3 Classification of the identified poles

The identification returns poles in three separate regions: a group between 1.05 and 1.33 Hz, a set between 14 and 24 Hz, and a set between 57 and 700 Hz that appears only when the records are analysed at the acquisition rate. Supplementary Table S2 lists the poles below 25 Hz by setup. Figure 5 shows why stabilisation alone cannot sort them: the 1.2 Hz pole and the 18–20 Hz cluster stabilise equally cleanly over all model orders.

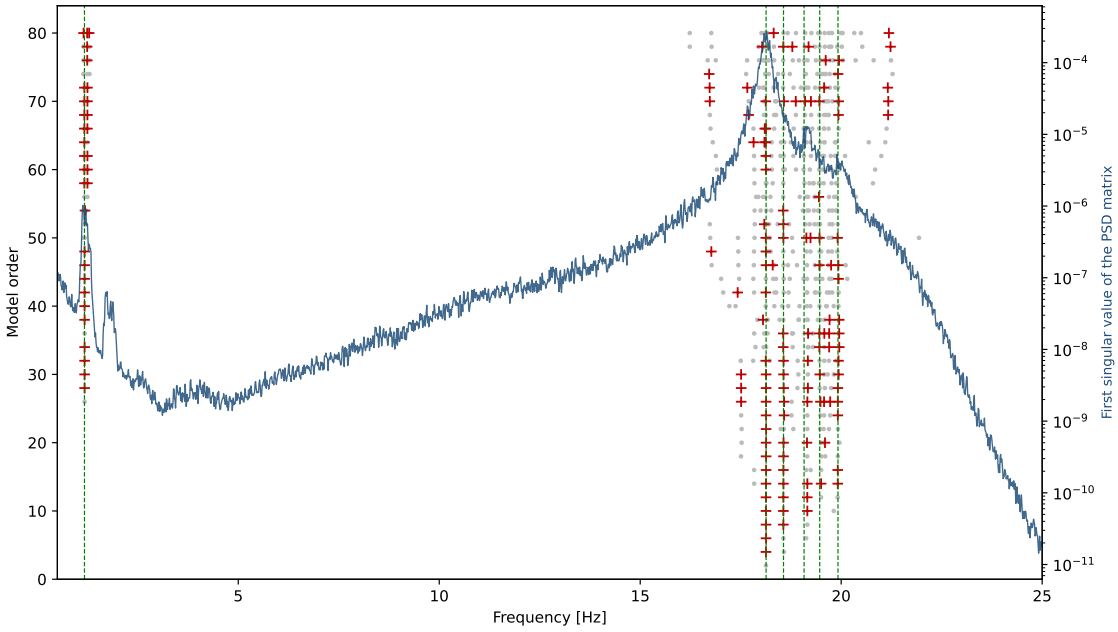


Figure 5: Stabilisation diagram for the vertical setup of campaign 1, with the first singular value of the cross-power spectral density matrix overlaid. Both the 1.2 Hz pole and the 18–20 Hz cluster stabilise cleanly over the full range of model orders, so the stabilisation behaviour alone does not distinguish them

The 1.05–1.33 Hz band passes all three checks. In this band five of the six setups to which the first check applies exceed its threshold, with effective channel counts from 1.55 to 5.00 (Figure 6). The exception is lateral setup 2 of campaign 1, the record with the least steady signal level; its low-band pole fails the check and is not used in any later comparison, although given that unsteady level the rejection may be a false one.

The band also passes the second check. Every setup returns poles within it, and it spans 13 % of its centre in campaign 1 and 16 % in campaign 2, within the width the model allows. In the vertical setups opposite corners of the frame move in antiphase, and in the lateral setups that pass the first check all horizontal channels move in phase with similar amplitude (Figure 7; Supplementary Table S3). Where the two vertical setups of campaign 2 measure the same five points, the best-matching pair of poles (1.147 and 1.074 Hz) has a modal assurance criterion (MAC) of 0.97. The individual members of the band are not resolved from one setup to the next (Supplementary Section S3).

The damping ratios, 1.3 to 9.9 %, are far above the bound of the third check. Re-identifying each record in three equal segments moves a pole by up to about 5 %, which gives the scatter of a single estimate.

The 14–24 Hz poles are local responses. In this band five of the seven setups have an effective channel count of 1.17 or less, four of them exactly 1.00: one channel responds and the others record only noise (Figure 6). They do not form one group either: in campaign 1 the

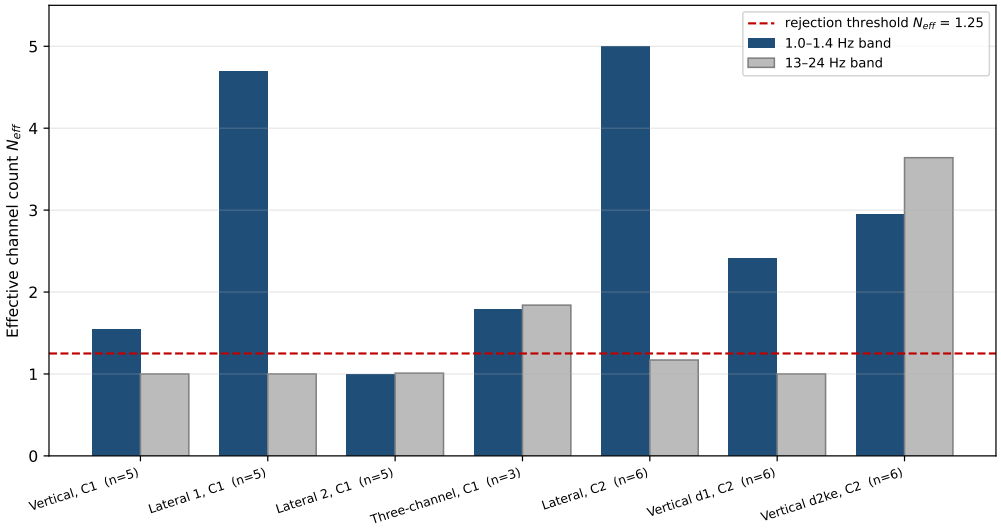


Figure 6: Effective channel count N_{eff} by setup and band, against the rejection threshold of 1.25 established by simulation. The number of channels in each setup is given in the axis label. A value of unity means that one channel carries the entire response

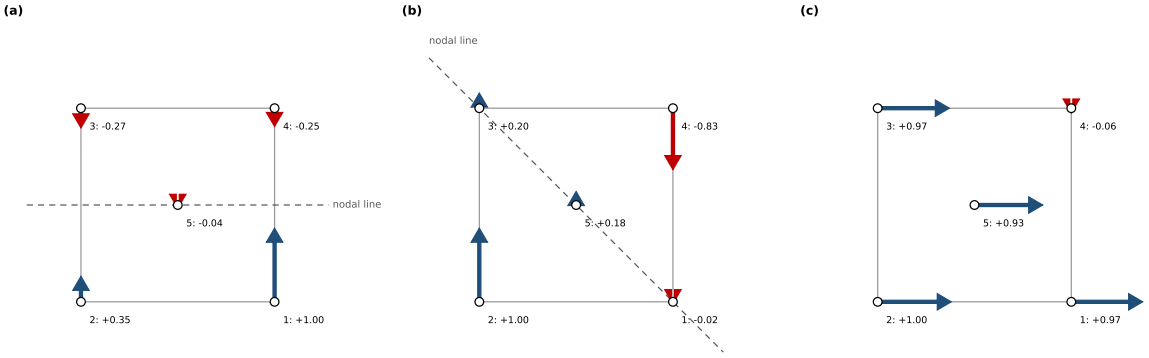


Figure 7: Measured deformation pattern of the 1.05–1.33 Hz band, taken from the first singular vector at the band peak. (a) Vertical setup of campaign 1. (b) Vertical setup d2ke of campaign 2. (c) Lateral setup of campaign 2. Signed amplitudes are given beside each node. The vertical setups show opposite corners in antiphase about a nodal line; the lateral setup shows the horizontal channels in phase, channel 4 being the only vertically oriented channel there. Vertical setup d1 of campaign 2 is given in Supplementary Table S3

vertical, the lateral and the three-channel setups return three different bands, 18.1–19.9 Hz, 21.6–23.8 Hz and 14.4–15.0 Hz, and in campaign 2 two of the three setups return no such pole. These poles may come from the bending of individual piles, whose first mode lies between about 8 and 16 Hz, or from the sensor mountings [40]; the records cannot tell which.

The poles at the acquisition rate include mains harmonics. Seven of the twenty poles returned between 57 and 700 Hz have damping between 0.007 and 0.042 %, far below the 0.1 % bound, and lie within 0.2 % of a multiple of 50 Hz (Supplementary Section S4). They enter through the measuring chain [55], but they look like structural poles to the estimator. The other thirteen were not classified, because the processing that returns them cannot resolve the structural band.

Table 4 summarises the result. A separate search below 0.5 Hz found no pole that passes the checks, only a spectral peak near 0.32 Hz in campaign 2, where the first tower mode is expected (Supplementary Section S5). A pole at 1.61–1.73 Hz appears in the three setups with horizontal channels other than lateral setup 2, which failed the first check; it lies outside the group and is left unclassified (Supplementary Section S3).

Table 4: Classification of the three groups of poles; n is the number of channels

Group	Check 1: N_{eff}/n	Check 2: same in every setup	Check 3: damping	Verdict
1.05–1.33 Hz	0.31–0.94 in five of six	yes, as a group; MAC 0.97 for the best-matching pair	1.3–9.9 %	structural
14–24 Hz	$N_{\text{eff}} \leq 1.17$ in five of seven	no; three different bands in one campaign, none in two setups of the other	0.3–6.0 %	local; not a mode of the cap
57–700 Hz	—	—	0.007–0.042 % (7 of 20 poles)	mains harmonics

5 Results and discussion

The geometric criterion uses the geometry of the foundation, for a cap whose mass lies near the pile heads. The identification uses the recorded motion, with one input from the model, the width allowed in the second check. The quantity compared below, the ratio of vertical to horizontal motion, plays no part in the checks.

5.1 The fundamental mode is a sway, not a rocking

The vertical setups show that the band involves a rotation about a horizontal axis: opposite corners of the frame move in antiphase about a nodal line (Figure 7). This pattern alone does not separate sway from rocking, since both rotate the cap. Only the height of the centre of rotation does.

The lateral setup of campaign 2 bounds that height from below. In this setup channel 4 measures vertically and the other cap channels horizontally, so the ratio of the vertical to the horizontal amplitude is the quantity bounded by (2). Table 5 gives it from several estimators: 0.08 to 0.16. Rocking predicts a ratio of 0.8 to 1.6 (Table 3), so the measured ratio is at least 5 times too small for rocking (Figure 8). It lies at or slightly above the sway bound of 0.09. The vertical signal of channel 4 is weak: its coherence with the central channel is 0.5, from about five spectral averages, which is not significant, so the ratio is an upper bound set by noise and by the transverse sensitivity of the accelerometer rather than a measured rotation. The data therefore

exclude rocking and are compatible with sway; they do not locate the centre of rotation at the focal point. Rocking could produce so small a ratio only if the cap happened to move within about $\pm 6^\circ$ of the one direction that places channel 4 on the rotation axis.

Two limits apply. The test rests on one setup, the only one whose channel orientations are recorded. And the bound was derived for the cap without the tower, whereas this setup was recorded with the tower standing; the tower may lower the centre of rotation and contribute to the excess.

Table 5: Vertical-to-horizontal amplitude ratio in the lateral setup of campaign 2, against the prediction of the pile-layout model for a channel 2.5 to 3.5 m from the rotation axis. The horizontal amplitude is the mean of channels 1, 2, 3 and 5

Estimator	f (Hz)	Vertical / horizontal
First singular vector at the band peak	1.203	0.084
Power spectral density ratio at the peak	1.203	0.116
the same, coherence-weighted (coherence 0.50)	1.203	0.082
Shape of the subspace pole	1.215	0.160
Predicted, sway: geometric bound (2)		≤ 0.09
Predicted, sway: full model		0.017–0.080
Predicted, rocking pair		0.8–1.6

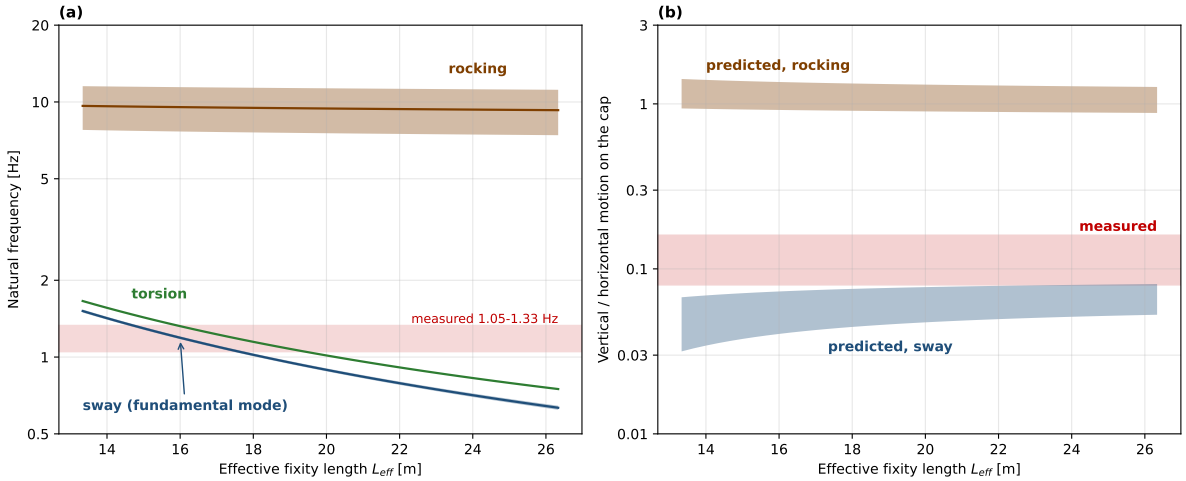


Figure 8: Predictions of the rigid-cap model of the T18 group against the effective fixity length L_{eff}, compared with the measurement (red strip). Each coloured band covers the axial pile stiffnesses estimated from the borehole, 2.5 to 5.8×10^8 N/m (Appendix B). (a) Natural frequencies: sway (the fundamental mode) and torsion fall in the measured range, rocking lies far above. (b) Vertical-to-horizontal motion at the cap top for a sensor 2.5 to 3.5 m from the rotation axis: the measured ratio lies at or slightly above the sway prediction and far below rocking

The same setup returns a second pole at 1.052 Hz, in which the channels on one edge of the frame move in antiphase with those on the opposite edge, as in a rotation about a vertical axis. The assignment is not secure. The central channel still carries 46 % of the corner amplitude, where torsion about the cap axis would give none, and the model places torsion 8 to 21 % above the fundamental pair for every combination of parameters, whereas this pole lies 13 % below the 1.215 Hz pole. The torsional mode is therefore not identified, and the pole is referred to below by its frequency.

The rocking mode was not identified either. The model places it between about 7 and 12 Hz for the axial stiffness estimated from the borehole (Appendix B), and no pole in that range

recurs in a second setup. Whether it is weakly excited or masked cannot be decided from these records.

5.2 Depth to fixity of the piles

Because the fundamental mode is resisted mainly by pile bending, its frequency depends on how deep below the seabed the piles are effectively fixed, and hardly on their axial stiffness. The measured frequency can therefore be inverted for the fixity length once the axial stiffness is bounded, here from the borehole (Appendix B). Inverting the setup means of campaign 1, 1.17 to 1.28 Hz, gives an effective fixity length of 14.2 to 17.0 m. Since 12.34 m of it is free length in water, the piles are effectively fixed 1.9 to 4.6 diameters below the seabed. An estimate that uses no measured frequency, the rule of Davisson and Robinson [56], gives 2.3 to 3.0 diameters, inside this range. The inverted value is an equivalent length for the group: it absorbs the interaction between piles, any flexibility of the pile–cap connection and the section assumed for the piles. Given also the uncertainty of the soil properties at $N \leq 2$, the comparison is a plausibility check, not a validation.

5.3 Effect of tower installation

Campaign 2 was recorded after the tower was erected. To see whether the band changed, the two campaigns must be compared between setups that measure in the same direction, because setups in different directions weight different members of the band (Figure 9; Supplementary Section S6).

Compared this way, the setup means differ by -2.3 and $-0.5\,\%$ in the vertical setups and by $-2.0\,\%$ for the 1.215 Hz pole of the lateral setup. These differences are smaller than the uncertainty of a single estimate. A pole moves by up to about $5\,\%$ between segments of one record, the best-matching poles of the two vertical setups of campaign 2 differ by $6\,\%$, and the spectral peaks differ from the subspace poles by up to about $5\,\%$ (Supplementary Section S6). A change of the fundamental frequency smaller than about $5\,\%$ can therefore be neither detected nor excluded. Two further limits apply. Environmental data, including the tide level, which changes the free length of the piles, were not recorded with either campaign. And the 0.32 Hz peak of campaign 2 suggests that the tower does contribute to the cap response at its own frequency.

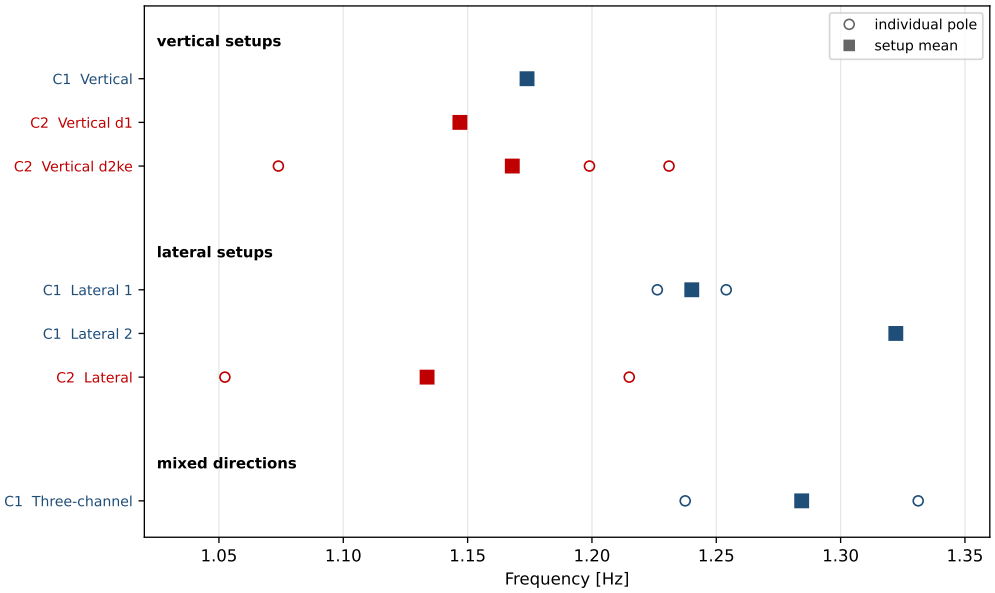


Figure 9: Poles of the 1.05–1.33 Hz band by setup, grouped by measured direction, with setup means. Blue: campaign 1, before the tower; red: campaign 2, after it. The means of like-for-like setups differ by 0.5 to 2.3 %. The larger difference between the lateral setups comes from the 1.052 Hz pole of campaign 2

6 Conclusions

This paper has shown how to tell sway from rocking in the fundamental mode of an elevated cap on inclined piles without knowing the stiffness of the soil, and how to decide from the records, with one input from the model, which identified poles are modes.

- When the piles are raked outward, their axes meet close to one point high above the cap, and the axial stiffness of the piles does not resist rotation about that point. The fundamental mode is therefore a sway about a centre at or above that point, resisted mainly by pile bending, and its vertical motion at the cap is bounded by the pile layout, for a cap whose mass lies near the pile heads. Rocking moves the cap vertically about as much as horizontally. The ratio of vertical to horizontal motion measured on the cap tells the two apart.

- Three checks based on the definition of a mode — it moves several sensors, it is the same in every setup, and it loses energy — showed that most of the poles that could be classified in the records of the T18 foundation are electrical interference or local responses, and that only the band near 1.2 Hz is a mode of the foundation.

- The vertical motion of that band is at most a tenth to a sixth of its horizontal motion. This excludes rocking and is compatible with sway; the single vertical channel available cannot place the centre of rotation more closely. The frequency of the band places the effective fixity of the piles two to five diameters below the seabed.

- Compared between setups measuring in the same direction, no change of this frequency larger than about 5 % could be resolved after the tower was installed.

These results come from a modest campaign: five or six accelerometers, records of 9 to 26 minutes, no environmental data, and records disturbed by electrical interference, knocks on the sensors and unsteady excitation. A short campaign of this kind, with a few horizontal channels on the cap and at least one vertical channel away from its centre, is therefore already useful, provided its poles are checked before they are interpreted.

A The rigid-cap model

Figure A.1 shows the model and its notation. The cap is a rigid body with six degrees of freedom, collected in $\mathbf{q} = (\mathbf{U}, \mathbf{\Phi})$: a translation $\mathbf{U}$ and a rotation $\mathbf{\Phi}$ about the origin O, taken at the centre of the cap soffit, where the pile heads are. Pile i has its head at $\mathbf{p}_i$, at radius r_i and azimuth θ_i, and is inclined outward at α_i from vertical. Its axis has the unit vector $\mathbf{a}_i = (\sin\alpha_i \cos\theta_i,\ \sin\alpha_i \sin\theta_i,\ -\cos\alpha_i)$, pointing from the head to the toe. The head is rigidly connected to the cap, so it displaces by $\mathbf{U} + \mathbf{\Phi} \times \mathbf{p}_i$ and rotates by $\mathbf{\Phi}$.

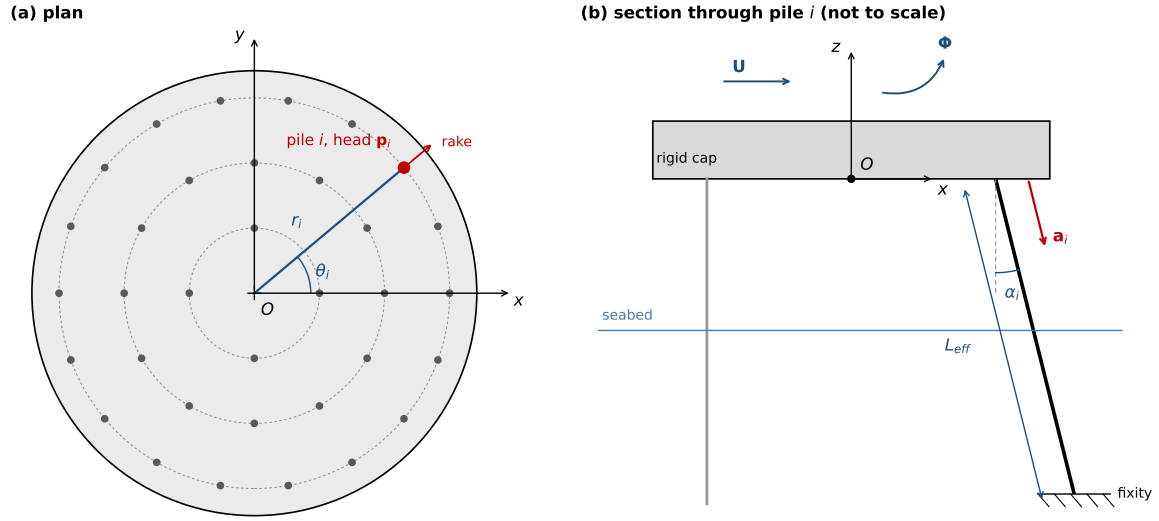


Figure A.1: Notation of the rigid-cap model. (a) Plan: the origin O at the centre of the cap soffit, and pile i with its head $\mathbf{p}_i$ at radius r_i and azimuth θ_i, raked radially outward. (b) Section through pile i, not to scale: the cap translation $\mathbf{U}$ and rotation $\mathbf{\Phi}$, the pile inclination α_i, the unit vector $\mathbf{a}_i$ along the pile axis, and the effective fixity length L_{eff}

Stiffness. Each pile is represented by an axial spring k_{ax} along $\mathbf{a}_i$ and by a beam of bending stiffness EI and torsional stiffness GJ, fixed at an effective fixity length L_{eff} measured along its axis. The beam gives the head the transverse stiffness $12EI/L_{\text{eff}}^3$, the rotational stiffness $4EI/L_{\text{eff}}$ and the coupling $6EI/L_{\text{eff}}^2$ between them. Writing $\mathbf{k}_i$ for this 6×6 head stiffness in the axes of the pile and $\mathbf{\Lambda}_i$ for the rotation from global to pile axes,

$$\mathbf{K} = \sum_i \mathbf{B}_i^{\mathsf{T}}\, \mathbf{k}_i\, \mathbf{B}_i, \qquad \mathbf{B}_i = \begin{bmatrix} \mathbf{\Lambda}_i & \mathbf{0} \\ \mathbf{0} & \mathbf{\Lambda}_i \end{bmatrix} \begin{bmatrix} \mathbf{I} & -\mathbf{S}(\mathbf{p}_i) \\ \mathbf{0} & \mathbf{I} \end{bmatrix}, \tag{A.1}$$

in which $\mathbf{S}(\mathbf{p})$ is the skew matrix of the cross product. Nothing in (A.1) decouples translation from rotation; all cross terms are retained.

Mass. The mass matrix is that of the rigid cap, 3.30×10^6 kg with its centroid 2 m above O, together with a participating share of the piles and of the water they carry, lumped at the pile heads. The central value of that share is 13/35 of the mass of each pile over L_{eff} and of the added water over the free length, the equivalent mass of a beam fixed at one end and guided at the other; the bounds 0 and 1 are carried through every result.

The axial part and the focal point. The axial springs contribute $k_{\text{ax}}\, \mathbf{g}_i \mathbf{g}_i^{\mathsf{T}}$ to $\mathbf{K}$, with $\mathbf{g}_i = (\mathbf{a}_i,\ \mathbf{p}_i \times \mathbf{a}_i)$, so their part of the stiffness is $k_{\text{ax}}\mathbf{G}$, where $\mathbf{G}$ depends only on the positions

and inclinations of the piles. For a translation u in x and a rotation φ about y,

$$G_{uu} = \sum_i \sin^2\alpha_i \cos^2\theta_i, \qquad G_{u\varphi} = \sum_i r_i \sin\alpha_i \cos\alpha_i \cos^2\theta_i,$$
$$G_{\varphi\varphi} = \sum_i r_i^2 \cos^2\alpha_i \cos^2\theta_i. \tag{A.2}$$

Pile i changes length by $\cos\alpha_i \cos\theta_i\,(u\tan\alpha_i + r_i\varphi)$, which gives the focal height (1) for a single ring. Table A.1 lists the values for the T18 group. The axial rotational stiffness is least about the centre at height $z^* = G_{u\varphi}/G_{uu} = 44.2$ m, where it is $G_{\varphi\varphi} - G_{u\varphi}^2/G_{uu} = 18.8$ m^2, 2 % of its value $G_{\varphi\varphi}$ about the pile heads. Almost all of this remainder comes from the vertical inner ring. Translation and rotation are strongly coupled, with a coupling coefficient of 0.990, and cannot be treated separately.

Table A.1: The axial part of the stiffness of the T18 group per unit k_{ax}, which depends on geometry alone

G_{uu}, $G_{u\varphi}$, $G_{\varphi\varphi}$, Eq. (A.2)	0.466 ; 20.6 m ; 930.6 m^2
Coupling coefficient $G_{u\varphi}/\sqrt{G_{uu}G_{\varphi\varphi}}$	0.990
Focal height $r\cot\alpha$, middle and outer ring	42.0 m ; 45.0 m
Centre of least axial rotational stiffness, $G_{u\varphi}/G_{uu}$	44.2 m
Rotational stiffness remaining about that centre	18.8 m^2 (2.0 %)

Parameter sweep. The modes of Table 3 are the eigenmodes of $\mathbf{K}$ and the mass matrix over all combinations of three uncertain inputs: the axial pile stiffness, from 1.5 to 6×10^8 N/m per pile; the depth to fixity, from 1.5 to 12 pile diameters below the seabed; and the participating pile mass, between its bounds 0 and 1. Over this range a fourfold increase in k_{ax} raises the fundamental frequency by only 5 to 6 %, and the centre of rotation of the fundamental pair stays between 48 and 150 m above the pile heads. The inequality $z_c \geq z_f$ used in (2) is thus a property of this model, in which the mass lies near the pile heads, and not a theorem.

B Axial pile stiffness and depth to fixity

The inversion of Section 5.2 needs a bound on the axial stiffness of the piles. It comes from the borehole. The shear modulus of each layer follows from its SPT value through the correlation of Ohsaki and Iwasaki [57], taken between 0.5 and 1.0 times its small-strain value G_0. The solution of Randolph and Wroth [10] for a compressible pile, with the interaction between piles [11], then gives an axial stiffness between 2.5 and 5.8×10^8 N/m per pile, a range over which the fundamental frequency changes by only about 3 %.

The effective fixity length is then the value of L_{eff} at which the fundamental frequency of the model of Appendix A equals the measured one. The measured values are the setup means of campaign 1, 1.17 to 1.28 Hz, without lateral setup 2, which failed the first check. The forward estimate uses the rule of Davisson and Robinson [56] for a pile in soil of constant modulus, with the modulus of the upper 3 to 5 m of the profile. Table B.1 collects the values.

Table B.1: Axial pile stiffness used as input, and the depth to fixity inverted from the measured frequency and estimated forward from the soil profile

Axial stiffness, isolated pile, $G = 0.5$–$1.0\,G_0$	4.7–5.8×10^8 N/m
Axial stiffness, in the group, interaction on the embedded length	2.5–3.4×10^8 N/m
Change of f_1 over 2.5–5.8×10^8 N/m	about 3 %
Depth to fixity, inverted from $f = 1.17$–1.28 Hz	1.9–4.6 pile diameters
at the campaign 1 mean, 1.233 Hz	2.3–4.0 pile diameters
Depth to fixity, Davisson and Robinson, $E_s = 3G$, $G = 0.5$–$1.0\,G_0$	2.3–3.0 pile diameters

Statements and Declarations

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Conflicts of Interest The authors declare no conflicts of interest.

Author Contributions Nguyen Hoang Bui: conceptualization, methodology, software, formal analysis, investigation, visualization, writing – original draft. Thi Cam Nhung Nguyen: conceptualization, methodology, investigation, resources, data curation, project administration, supervision, writing – review and editing. Huu-Quyet Nguyen: investigation, data curation. Chung Truong Thanh: software, validation. Hung Nguyen-Dang: writing – review and editing. José C. Matos: writing – review and editing. All authors read and approved the final manuscript.

Data Availability Statement The acceleration records analysed in this study are openly available in Zenodo at `https://doi.org/10.5281/zenodo.23177722` [58]. The design drawings belong to the project owner and are not redistributed; every quantity read from them is given in Table 1 and Figure 2.

Code Availability The analysis is implemented in Python (NumPy, SciPy, Matplotlib). The code that regenerates every reported quantity, table and figure from the records is publicly available at `https://github.com/chungntu/t18-foundation-oma`.

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